When Order Volume Masks Operational Concentration Risk

Executive Summary

This analysis was built using a PostgreSQL reporting environment connected to Power BI to examine operational sales activity, cancellation exposure, and warehouse-related demand patterns across a multi-year order dataset.

The analysis identified a significant gap between headline order activity and realised operational value. While total ordered revenue reached approximately £79.0M, only £50.2M translated into active revenue, with 36.4% of ordered revenue linked to cancelled orders. Profitability was also materially affected, with 28.8% of total projected profit associated with cancelled activity. Operationally, the impact extends beyond financial reporting. Approximately 35K units were ordered during the analysed period, of which around 9K units were associated with cancelled orders — representing approximately 25.7% of stock prepared or allocated for demand that was ultimately not fulfilled.

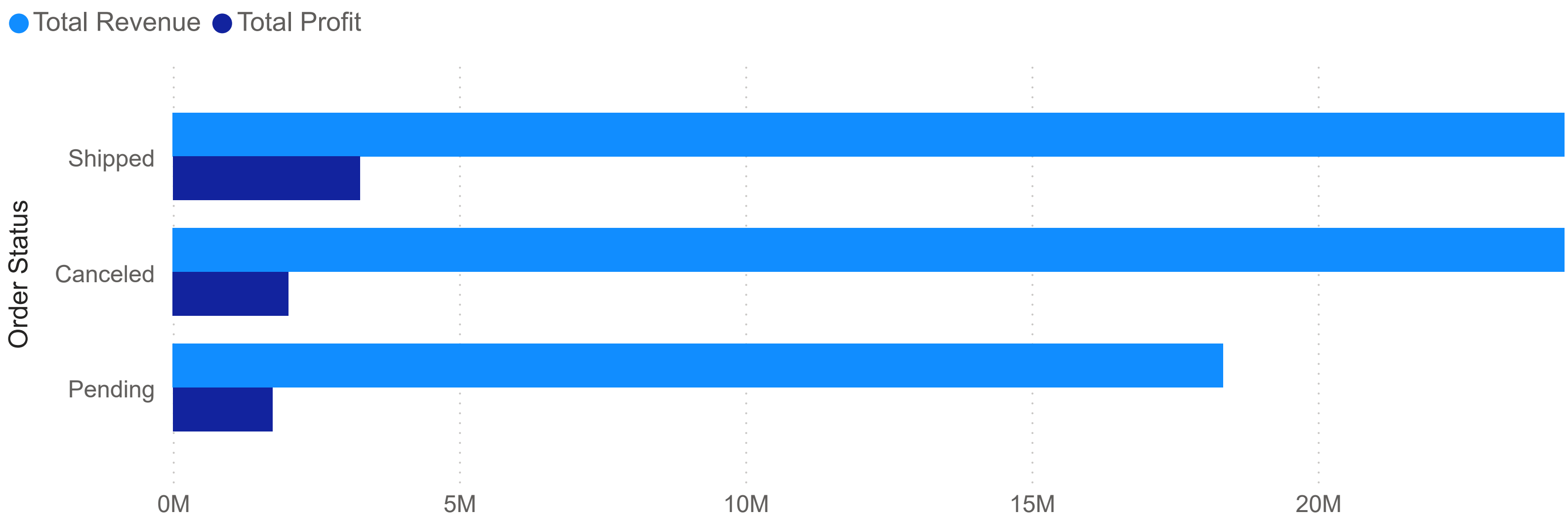
This creates several operational pressures:

- warehouse space consumed by unstable demand
- procurement and stock allocation inefficiencies
- administrative overhead from cancellations and reallocation
- reduced visibility of true commercial performance
- increased operational volatility across key product categories

The analysis also highlighted product-category concentration within both actual and cancelled demand. Storage and CPU-related categories generated the highest operational volumes, while also contributing materially to cancellation exposure.
The current dataset provides strong operational visibility at order and product level. However, limitations in location and fulfilment linkage reduce the ability to analyse cancellations regionally or evaluate warehouse fulfilment efficiency against customer geography. Strengthening the integration between customer location data, warehouse allocation, and fulfilment activity would significantly improve operational forecasting and distribution insight.

Revenue Ordered	Actual Revenue	% Cancelled	Profit Ordered	Actual Profit	% Cancelled
79.00M	50.23M	36.41%	7.03M	5.01M	28.76%

Total Revenue and Total Profit by Order Status



Source: [INVENTORY MANAGEMENT ANALYSIS (SQL BASED PROJECT)]

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Recommendations

- **Introduce Cancellation Exposure Monitoring**

Implement regular monitoring of cancellation percentages across revenue, profit, and quantity metrics at both operational and product-category level. Current cancellation levels indicate a material gap between demand captured and demand fulfilled, creating financial distortion and operational waste.

- **Review Stock Allocation and Procurement Strategy**

Approximately one quarter of ordered quantity did not convert into shipped activity. This suggests procurement and warehouse resources may be allocated toward unstable demand patterns. Product categories with consistently high cancellation exposure should be reviewed for purchasing thresholds, stock holding policies, and order validation controls.

- **Develop Exception-Based Operational Reporting**

Several unusually large orders and cancellation events materially influence operational totals and planning assumptions. Introducing exception monitoring for high-value orders, large quantity spikes, and cancellation anomalies would improve operational visibility and reduce exposure to concentration risk.

- **Improve Warehouse and Regional Data Integration**

The current dataset does not directly link order lines to warehouse fulfilment locations, and customer address data cannot currently be matched reliably to warehouse regions. Enhancing these relationships would enable regional cancellation analysis, fulfilment optimisation, and more accurate warehouse-capacity planning.

- **Use Demand Trends to Support Expansion and Capacity Planning**

If customer demand increases consistently within particular geographic areas, additional warehouse locations or revised fulfilment strategies may reduce shipping delays, fulfilment cost, and cancellation risk. Improved regional visibility would allow operational growth decisions to be supported by measurable demand patterns rather than reactive expansion.

Total Cancelled Revenue & Actual Profit by Product Category